

Experiment No. 5

Environment: Microsoft Windows

Tools/ Language: Oracle

OBJECTIVE: To implement the concept of aggregating & grouping of Data. Theory:

Group Functions:

Group functions operate on set of rows, result is based on group of rows rather than one result per row as returned by single row functions.

1) **Avg** return average value of n

Syntax:

Avg ([distinct/all] n)

2) **Min** return minimum value of expr.

Syntax:

MIN([distinct/all] expr)

3) **Count** Returns the no of rows where expr is not null

Syntax:

Count ([distinct/all] expr)

Count (*) Returns the no rows in the table, including duplicates and those with nulls.

4) **Max** Return max value of expr

Syntax:

Max ([distinct/all]expr)

5) **Sum** Returns sum of values of n

Syntax:

Sum ([distinct/all]n)

Example Queries:

Aggregate Functions Examples:

select AVG(GPA) as "Average GPA" from student;

Average GPA

3.56666667

Select AVG(DISTINCT GPA) as "Distinct GPA" from student

Distinct GPA

3.54285714

Select MIN(GPA) as "Minimum GPA" from student

Minimum GPA

2.9

Select COUNT(*) from student
COUNT(*)

12

Select COUNT(sName) from student
COUNT(SNAME)

12

select COUNT(distinct sName) from student
COUNT(DISTINCTSNAME)

10

Select MAX(GPA) from student
MAX(GPA)

3.9

Select SUM(GPA) from student
SUM(GPA)

42.8

Q.Find average GPA without using Average function.

Select SUM(GPA)/COUNT(GPA) as “Average” from student
Average

3.56666667

Grouping Data From Tables:

There are circumstances where we would like to apply the aggregate function not only to a single set of tuples, but also to a group of sets of tuples, we specify this wish in SQL using the group by clause. The attribute or attributes given in the group by clause are used to form group. Tuples with the same value on all attributes in the group by clause are placed in one group.

Syntax:

SELECT columnname, columnname
FROM tablename
GROUP BY columnname;

HAVING clause:

The HAVING clause can be used in conjunction with the GROUP BY clause. HAVING imposes a condition on the GROUP BY clause, this further filters the groups created by the GROUP BY clause.

HAVING clause can be used to find duplicates in a relation or in other words find unique values in the situations where DISTINCT cannot apply.

NOTE: Each column specifications specified in HAVING clause must occur in the list of columns named in the GROUP BY clause.

Example

1. select state, count(cname) from college GROUP BY state;

STATE	COUNT(CNAME)
-----	-----
CA	2
MA	1
NY	1

2. select state, count(cname) from college GROUP BY state having count(cname)>1;

STATE	COUNT(CNAME)
-----	-----
CA	2

Practical Assignment - 5

Department: Computer Engineering & Applications

Course: BCA

Subject: Database Management System Lab (BCAC0816)

Year: 1ST

Semester: 2ND



Create these tables which consist of following attributes

College

Column Name	Data type	Size
cName	varchar2	10
cstate	varchar2	10
enrollment	int	

Student

Column Name	Data type	Size
sID	int	
sName	varchar2	10
CGPA	number	2,1
marks	int	
DoB	date	

Apply

Column Name	Data type	Size
sID	int	
cName	varchar2	10
major	varchar2	20
decision	char	1

Insert the following data to these tables:

Student

sID	sName	CGPA	Marks
123	Amit	3.9	1000
234	Balbir	3.6	1500
345	chirag	3.5	500
456	Dev	3.9	1000
567	Eshan	2.9	2000
678	Faizal	3.8	200
789	Garvit	3.4	800
987	Himani	3.7	800
876	Ishan	3.9	400
765	Jay	2.9	1500
654	Amit	3.9	1000
543	chirag	3.4	2000

College

cName	state	enrollment
Stanford	CA	15000
Berkeley	CA	36000
MIT	MA	10000
Cornell	NY	21000
Harvard	MA	50040

Apply

sID	cName	major	decision
123	Stanford	CS	Y
123	Stanford	EE	N
123	Berkeley	CS	Y
123	Cornell	EE	Y
234	Berkeley	biology	N
345	MIT	bioengineering	Y
345	Cornell	bioengineering	N
345	Cornell	CS	Y
345	Cornell	EE	N
678	Stanford	history	Y
987	Stanford	CS	Y
987	Berkeley	CS	Y
876	Stanford	CS	N
876	MIT	biology	Y
876	MIT	marine biology	N
765	Stanford	history	Y
765	Cornell	history	N
765	Cornell	psychology	Y
543	MIT	CS	N

State the SQL Queries for each of the following:

- Q1. Count the total number of Students.
- Q2. Calculate the average CGPA of all Student.
- Q3. Determine the minimum and maximum GPA. Rename the titles as 'max_CGPA' and 'min_CGPA' respectively.
- Q4. Count the number of students having CGPA greater than or equal to 3.7.
- Q5. Find Maximum, Average, Minimum, total CGPA of all student.
- Q6. Find total number of colleges in our Application Database.
- Q7. Find how many different majors student had applied in.
- Q8. Find total no. of Applications in our Application System's Database.
- Q9. Find average of all distinct CGPA.
- Q10. Display the total number of application accepted.
- Q11. Find number of students having CGPA>3.4 and coming from high school having marks>1000.
- Q12. Find how many students applied to 'marine biology'.
- Q13. Find how many applications were rejected and accepted by the colleges.
- Q14. Give state and number of colleges of a state that has more than 1 college.
- Q15. Find the name of students that are duplicate.